

Changyue (William) Zhao

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EDUCATION

Duke University, Durham, NC

08/2025 – Present

Master of Biostatistics

- GPA: 4.0/4.0
- Selected Coursework: *Statistical Methods for Learning and Discovery, Deep Learning for Health Data, High-Dimensional Statistics and Machine Learning, Real Analysis, Statistical Inference (Ph.D. Level)*
- Ranked second in Real Analysis by course grade (Instructor: Michael C. Reed)

University of Minnesota, Twin Cities, Minneapolis, MN

09/2023 – 05/2025

Bachelor of Science in Statistical Science, Minor in Computer Science

- GPA: 4.0/4.0
- Selected Coursework: *Mathematical Analysis, Theory of Statistics, Regression and Correlated Data, Statistical Machine Learning, Applied Multivariate Methods, Algorithms and Data Structures, Graduate Data Mining*
- Graduated with High Distinction and received the Buehler Memorial Undergraduate Award as one of eight nominees. Named to the CLA Dean's List from Fall 2023 to Spring 2025

Shandong University of Finance and Economics, Jinan, China

09/2021 – 06/2025

Bachelor of Economics in Finance

- GPA: 92.86/100 (WES GPA: 3.89/4.0)
- Selected Coursework: *Advanced Mathematics, Linear Algebra, Probability Theory and Statistics, Applied Statistics, Python, Microeconomics, Finance*
- Received the SDUFE First-Prize Scholarship in 2022 and 2023

RESEARCH INTERESTS

- Applications of foundation models and LLM-based agents for biomedical analysis
- Deep learning methods for biomedical data analysis and multimodal integration

PUBLICATIONS

Changyue Zhao and Zhicheng Ji.

09/2025 – Present

Visual LLM-guided consensus spatial domain detection with L-STAR.

First-author manuscript accepted for publication in Nature Communications

- Developed the L-STAR visual LLM workflow for spatial domain detection, with structured prompts, repeated comparisons of spatial maps and optional histology, JSON logs, and win-rate-based method selection.
- Implemented consensus clustering with co-assignment matrices and average linkage. Added an all-wise comparison mode that aggregates repeated rankings to select ensemble members with one LLM call per repeat.
- Designed comparative experiments across six spatial transcriptomics datasets to examine ensemble size, alternative LLMs, and fixed-method consensus baselines, separating the roles of method selection and clustering aggregation.
- Built marker-gene-based domain annotation with spot- and cell-level labels and an Unknown option. Evaluated agreement with manual annotations using BioLORD-2023 embeddings and within-dataset permutation tests.
- Audited sensitivity to color palettes, presentation order, and redundant candidate methods by examining ranking agreement, selected-method overlap, and clustering stability through repeated runs and controlled perturbations.
- **Preprint:** [bioRxiv](#) **Open-source software:** [L-STAR](#) **Supervisor:** Prof. Zhicheng Ji, Duke University

Changyue Zhao and Zhicheng Ji.

01/2026 – Present

TRACE: Trajectory Reasoning via AI-guided Cross-view Evidence.

Manuscript in preparation

- Developed **trace**, an end-to-end **LangGraph** package coordinating Python, R/Seurat, and LLM inference for single-cell and spatial pseudotime analysis. Implemented schema validation, corrective retries, provenance tracking, and fingerprint-validated replay with support for expert-supplied inputs.
- Designed parallel UMAP and t-SNE workflows for cell-type annotation, lineage-family adjudication, and cross-embedding trajectory fusion, using assay-aware molecular evidence for scRNA-seq and scATAC-seq inputs.
- Designed LLM-guided spatial process-instance resolution using molecular profiles and tissue topology to distinguish repeated local processes within connected tissue, with explicit cell membership, local seeds, and unresolved assignments.
- Implemented expression-weighted graph pseudotime with instance-level ordering review and partial-trajectory support. Separated molecular lineage hypotheses, physical feasibility, and spatial evidence in auditable execution rules.

- Built an optional completion module with evidence-based assignment review and donor-restricted imputation. Preserved original inference outputs and unresolved multi-instance assignments, keeping graph-derived and imputed values distinct.
- Designed and implemented **bPOS**, an exact branch-aware pseudotime metric that jointly evaluates lineage membership and within-lineage ordering using scalable block processing and inclusion-exclusion counting.
- Diagnosed a bug in the **dyneval**-based benchmark evaluation of **F1branches**. Reimplemented the published Jaccard, recovery, relevance, and harmonic-mean definition over the full cell scope and added diagnostic comparisons.
- Built reproducible comparisons with Monocle 3, Slingshot, and TSCAN. Designed synthetic negative controls, perturbation tests, and staged replay experiments to assess spatial assumptions, instance partitioning, and pseudotime stability.

RESEARCH EXPERIENCE

Med2State

11/2025 – Present

School of Medicine, Duke University, Durham, NC

- Designing Med2State, a framework for learning structured patient-state representations from incomplete multimodal clinical evidence and longitudinal history, with separate disease, procedure, and treatment channels.
- Designing configurable encoders for demographics, measurements, structured EHR events, text, and images. Exploring semantic attention and low-rank fusion with temporal updates that account for irregular visit intervals.
- Formulating evidence-subset and longitudinal learning objectives with typed concept readouts, using only information available at each state time to preserve clinical semantics and prevent future-data leakage.
- Planning controlled simulations, component ablations, and clinical-record evaluations of concept readout, similar-visit retrieval, and trajectory analysis, with stress tests for missing inputs and model misspecification.
- **Supervisor:** Prof. Chuan Hong, Duke University

PaperEval

03/2025 – 09/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Built **Selenium** and asynchronous data-collection pipelines for major machine-learning conferences on OpenReview, storing papers, reviews, scores, author responses, and metadata in JSON with resumable checkpoints.
- Extracted author contact information from conference PDFs with PyMuPDF to enrich OpenReview metadata.
- Fine-tuned GPT-4o for acceptance prediction using progressively richer inputs: paper metadata, reviewer scores, and review comments. Implemented training-data preparation and classification evaluation workflows.
- Developed LLM-based agent workflows for manuscript and figure assessment, review-response scoring, and response generation. Evaluated generated responses against author replies from accepted and rejected submissions.
- Designed a bounded Gaussian-noise mapping to convert acceptance levels into continuous constructiveness targets.
- **Open-source software:** [PaperEval](#)  **Supervisor:** Prof. Jie Ding, University of Minnesota

MUDAC (Midwest Undergraduate Data Analysis Competition) 2025

03/2025 – 04/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Encoded sparse, high-dimensional numerical and text data with embeddings, sentiment scores, and pairwise similarities.
- Applied Cox proportional hazards regression and a causal Bayesian network to study match duration and closure.
- Designed a mixture-of-experts model with regression, MLP, and Transformer experts for different feature types and a learned gating network. Compared it with random forest and XGBoost baselines.
- Placed 7th in Innovation & Creativity.
- **Supervisors:** Barbara Kuzmak and Robert Oliver VandenBurg, University of Minnesota

University of Minnesota & Hennepin County Public Health Equity Data Challenge

10/2024 – 11/2024

School of Public Health, University of Minnesota, Twin Cities, Minneapolis, MN

- Analyzed Hennepin County questionnaire data on inequities in access to mental-health resources using multiple regression, ridge regression, decision trees, and causal inference methods.
- Built a mental-health prediction interface in **tkinter** backed by an **AdaBoost** model in **scikit-learn**.
- Co-authored the analysis report and poster and placed in the top 25% of the undergraduate division.
- **Supervisors:** Marta Shore and Prof. Julian Wolfson, University of Minnesota

Overlapping Group LASSO Regression R Package Implementation

09/2024 – 05/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Compared **overlapping_LeastR** from the MATLAB SLEP library with **oglasso** in the R package **VennLasso** under matched problem sizes and convergence tolerances.
- Simulated 1,000 large sparse-matrix datasets with overlapping groups and compared runtime using t-tests.
- Implemented the optimized algorithm in C for integration with R.
- **Supervisor:** Prof. Tianxi Li, University of Minnesota

PROFESSIONAL EXPERIENCE

Teaching Assistant: BIOSSTAT 830

08/2026 – Present

Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

Supervisor: Dr. Lin (Lynn) Lin

- Hold weekly office hours and grade assignments in graduate statistical inference, asymptotic theory, and decision theory.
- Prepare and verify proof-based LaTeX assignment solutions. Format and proofread lecture slides, standardizing mathematical notation and converting formulas into editable PowerPoint equations.

Undergraduate Teaching Assistant: STAT 3011

09/2024 – 05/2025

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Taught weekly 50-minute sections for about 30 students on probability, sampling, hypothesis testing, regression, and R.
- Developed instructional slides, held office hours, and assessed weekly assignments and exam responses through Gradescope. Shared teaching materials with the course team.
- Teaching-language qualification: ELP 1, the highest rating on the Spoken English Test for Teaching Assistants (equivalent to TOEFL iBT Speaking 27–30/30 or IELTS Speaking 8.5–9.0/9).
- **Materials:** [GitHub](#)  **Supervisors:** Yuyoung Park, Aritra Banerjee, Kazeem Adepoju, University of Minnesota

OTHER INFORMATION

- **Programming:** Python, R, C/C++, Java, OCaml, Swift, MATLAB, SQL
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face
- **Scientific Computing:** Scanpy, Seurat, NumPy, pandas
- **Tools:** Git, SQLite, Selenium, L^AT_EX, Markdown, SPSS
- **Hobbies:** Photography, Snowboarding, Car Racing, Cycling, Powerlifting